



Computer Organization & Assembly Language

Course: Computer Organization & Assembly Language (Lab) - CS-318

2.1 Data Declaration

In 8086 assembly language, data is declared in the **DATA SEGMENT** using different data types.

Directive	Meaning	Size
DB	Define Byte	1 byte
DW	Define Word	2 bytes
DD	Define Double Word	4 bytes
DQ	Define Quad Word	8 bytes

Example:

```
DATA SEGMENT
```

```
A DB 10H
```

```
B DB 20H
```

```
C DW 1234H
```

```
MSG DB "HELLO$", 0
```

```
DATA ENDS
```

2.2 Direct Addressing

In **direct addressing**, the operand's **memory name** or **offset** is used directly in the instruction.

Example:

```
MOV AX, DATA
```

```
MOV DS, AX
```

```
MOV AL, A ; Move contents of variable A into AL
```

```
MOV BL, B ; Move contents of variable B into BL
```

```
ADD AL, BL
```

```
MOV RESULT, AL
```

Explanation:

The processor accesses data by directly using its memory label (e.g., A, B, etc.).

2.3 Size Mismatch Errors

A **size mismatch** occurs when the source and destination operands are of **different sizes**.

Example (Error):

```
MOV AL, RESULT ; RESULT is a word, AL is a byte → size mismatch
```

Correct Version:

```
MOV AX, RESULT ; Both are 16-bit
```

Explanation:

Always ensure that both operands have the **same data size** — either byte or word.

2.4 Register Indirect Addressing

In this mode, a **register** holds the **address** of the operand.

Registers allowed: BX, BP, SI, DI

MOV destination, [register]

means "use the address stored in the register"

Example:

MOV BX, NUM1 ; BX = address of NUM1

MOV AL, [BX] ; AL = value at address BX

2.5 Register + Offset Addressing

In this mode, an **offset** (constant value) is added to the **address stored in a register**.

Example:

MOV SI, OFFSET NUM1

MOV AL, [SI + 1] ; Access the next byte after NUM1

Explanation:

The final address is calculated by adding the value of SI and 1 (offset).

2.6 Segment Association

In 8086, memory is divided into **segments** (like DATA, CODE, STACK).

To access variables from the **DATA segment**, it must be associated with DS (Data Segment Register).

Example:

MOV AX, DATA

MOV DS, AX

This ensures that all variables like A, B, NUM1 belong to the **current data segment**.

Observation Table

Mode	Example	Explanation
Direct	MOV AL, A	Access memory label directly
Register Indirect	MOV AL, [BX]	Access data using address in register
Register + Offset	MOV AL, [SI + 1]	Access data by adding offset
Segment Association	MOV DS, AX	Connect data segment to DS register

Lab Exercise

Write a program in 8086 assembly language that uses **all addressing modes** learned in this lab:

1. **Direct Addressing**
2. **Register Indirect Addressing**
3. **Register + Offset Addressing**
4. **Segment Association**

Requirements:

- Use arithmetic instructions: ADD, SUB, MUL, DIV.
 - Store the results of each arithmetic operation in memory.
 - Display clear comments showing which addressing mode is used.
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